

Problem 1 – Another Sum. Prove that the sum of the first n odd numbers is n^2 . Namely, prove that for all $n \in \mathbb{N}$,

$$\sum_{i=1}^n (2i + 1) = (n + 1)^2.$$

Problem 2 – Induction Inequality. Prove that $n < 2^n$ for $n \in \mathbb{N}$.

Problem 3 – Find the error in this proof of $P(n)$.

$P(n)$: Every set of n distinct lines in a plane (where $n \geq 2$), no two of which are parallel, meet at the same point.

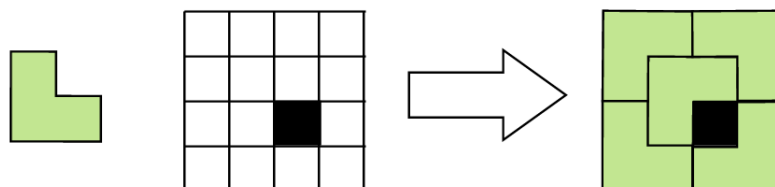
Proof: We proceed by induction on n .

Our base case is $P(2)$. Any 2 lines in a plane that are not parallel meet at a common point by definition of parallel lines, so $P(2)$ is true.

Next, let k be arbitrary in $\mathbb{N}$ and assume that $P(k)$ is true — every set of k distinct lines in a plane, no two of which are parallel, meet at a common point. We show that $P(k + 1)$ is true. Consider a set of $k + 1$ distinct lines in a plane, no two of which are parallel. By the inductive hypothesis, the first k of these lines meet at a common point; call this point p_1 . Similarly, the last k of these lines meet at a common point; call this point p_2 . Note that, apart from the first and last line in the plane, all other lines cross both p_1 and p_2 . We will show that p_1 and p_2 must be the same point, and thus all $k + 1$ lines meet at the same point.

Recall that two points define a line. If p_1 and p_2 were different points, all lines containing both p_1 and p_2 would be the same line. This contradicts our assumption that all lines are distinct. Thus, both p_1 and p_2 are the same point, and we conclude that this point is the intersection of all $k + 1$ lines. Therefore, we have shown that if $P(k)$ is true then $P(k + 1)$ is true, completing our proof. $\square$

Problem 4 Consider a square $N \times N$ grid where $N = 2^k$ is a power of 2, and imagine placing L -shaped domino pieces of area 3 on this grid. A **tiling** of the grid is a way of placing the pieces so that no two pieces overlap and all grid cells are covered. In the starting grid, one square is excluded, and doesn't need to be covered by any piece. You are given L -shaped domino pieces (in light green) and a grid, and you must tile the entire grid (excluding the black square). An example of a valid tiling is shown on the right.



Note: every domino piece has the shape described above and covers exactly three squares, every input grid has exactly one excluded (black) square, but the location of the black square may change between input grids.

Prove by induction on k that every $N \times N$ grid can be tiled successfully where $N = 2^k$ and the black square can exist anywhere on the grid.

Problem 5 – Water Balloon Fight. There are $2n + 1$ people at a water balloon fight, each with one water balloon. When the fight starts, each person throws the balloon at their nearest neighbor (assume, for the sake of the problem, that there are equally-near neighbors!). Show that, for $n > 0$, at least one person is not hit. (Hint: Start by looking at the two people closest to each other.)